

TalkTrace Project Modules & System Specifications

Comprehensive Breakdown of System Modules, Tech Stack, & Capabilities

SECTION 1: SYSTEM MODULE ARCHITECTURE OVERVIEW

TalkTrace is structured into 10 modular subsystems that manage the end-to-end meeting lifecycle: from automated calendar bot auto-joins and zero-trust stream encryption to GPU speech diarization, local LLM executive minutes generation, client-side vector search, and task board exports.

MODULE 1: Authentication & User Access Security Module

Handles user registration, login, email verification enforcement, and client-side RSA/AES key pair generation.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
1.1	Firebase Auth Engine	Firebase SDK v9	Handles email/password sign-up, sign-in, and auth tokens.
1.2	Email Verification Pass	Firebase ActionCode	Dispatches verification links on signup; blocks unverifi...
1.3	RSA Key Generator	node-forge	Generates 2048-bit client RSA key pairs in memory for ze...

MODULE 2: Zero-Trust Cryptographic Audio Vault Module

Protects audio files and transcript archives using zero-cloud stream encryption and crypto-shredding.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
2.1	AES-256-CBC Stream Cipher	Node.js crypto	Encrypts incoming audio streams directly to disk vault.
2.2	Key Blob Manager	SQLite3 / Crypto	Encrypts meeting keys using RSA-OAEP before DB storage.
2.3	Crypto-Shredding Engine	FS / SQLite	Permanently deletes encrypted audio and key blobs upon u...

MODULE 3: Audio Ingestion & Note-Taker Bot Automation Module

Deploys AI note-taker bots to Google Meet, Zoom, and Teams rooms to capture audio streams automatically.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
3.1	Recall.ai Bot Launcher	Axios / REST API	Launches headless audio recording bots into web meeting ...
3.2	FFmpeg Transcoder	fluent-ffmpeg	Converts raw audio streams to optimized 16kHz mono MP3 f...
3.3	Pipeline Lifecycle Manager	Node.js Streams	Orchestrates step-by-step processing states (downloaded ...

MODULE 4: Diarization & Speech Transcription Engine Module

Transcribes audio into text with word-level timestamps, Pyannote speaker diarization, and Google Translate language detection.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
4.1	WhisperX Engine	Python / PyTorch	High-speed Whisper speech-to-text transcription with ali...
4.2	Pyannote Diarizer	Pyannote 3.1	Differentiates multi-speaker tracks (SPEAKER_00 -> real ...
4.3	Google Translate Detection	Google GTX API	Keyless 99.8% precision language auto-detection for 14+ ...

MODULE 5: Local AI Executive Minutes & Action Item Extractor Module

Runs airgapped local LLMs to generate structured 6-part executive meeting minutes.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
5.1	Ollama Local LLM	Llama 3.2 / Mistral	Generates airgapped JSON summary without external cloud ...
5.2	Executive Minutes Generator	Node.js / Ollama	Extracts Agenda, Discussion Points, Decisions, Risks, Ac...
5.3	Priority Rating Normalizer	JSON Parser	Assigns High, Medium, Low priority ratings to action ite...

MODULE 6: Automated Task Manager & Exporters Module

Provides interactive task progress bars, completion checkboxes, and multi-platform project board exporters.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
6.1	Interactive Task Board	React Component	Checkable action items with progress completion percenta...
6.2	CSV & Jira Exporters	Blob / REST API	One-click exporter to CSV spreadsheets and Jira / GitHub...

MODULE 7: Calendar Integration & Auto-Join Scheduler Module

Syncs upcoming Google and Outlook calendar events, schedules bot auto-joins, and dispatches attendee report emails.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
7.1	Google & Outlook Sync	Google / Graph API	Syncs upcoming meeting links (meet.google.com, teams.mic...
7.2	Auto-Join Scheduler	Node.js Scheduler	Dispatches bot auto-joins 2 minutes prior to meeting sta...
7.3	Attendee Report Emailer	REST API	Sends executive summary email reports directly to meetin...

MODULE 8: Local Semantic Vector RAG Search Engine Module

Generates 128-d dense sentence vectors to search meeting archives using natural language queries.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
8.1	Sentence Vector Generator	JS Math Engine	Computes 128-dimensional dense vector embeddings for tex...
8.2	Cosine Similarity Ranker	Math Vector Dot	Calculates exact cosine similarity scores (% Vector Sim)...
8.3	IndexedDB Local Cache	HTML5 IndexedDB	Stores client-side vector embeddings for zero-latency lo...

MODULE 9: Speaker Analytics & Diarization Insights Engine Module

Computes talk-time percentages, speaking speeds (WPM), speech gaps, and sentiment breakdowns.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
9.1	Talk-Time Charting	React / Canvas	Visualizes speaker turn shares and participant talk-time...
9.2	WPM & Gap Metrics	Math Analytics	Measures words-per-minute pace and inter-speaker pause d...

MODULE 10: Hyper-Futuristic Cybernetic UI Showcase Module

Provides dark obsidian glassmorphism UI layouts and interactive spatial node canvas topology.

No	Subcomponent	Tech / Stack	Key Capabilities & Logic
10.1	CyberDashboard UI	React / Glassmorphism	Obsidian backdrop, glowing radial SVG orbit status rings...
10.2	NeuralLattice Canvas	React / HTML5 Canvas	Spatial node topology, floating speaker nodes, central A...